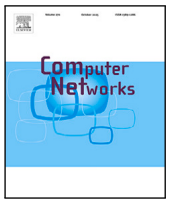
Computer Networks 283 (2026) 112320



Contents lists available at ScienceDirect

Computer Networks

journal homepage: www.elsevier.com/locate/comnet





XAI-CPS: An explainable hybrid deep learning framework for real-time anomaly detection and adaptive threat mitigation in networked Cyber–Physical Systems ☆

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ARTICLE INFO

Keywords:
Cyber–Physical Systems
Anomaly detection
Explainable AI
Deep learning
Intrusion detection
Reinforcement learning
Industrial IoT security

ABSTRACT

The increasing reliance of critical infrastructure on networked Cyber–Physical Systems (CPS) has exposed these systems to sophisticated multi-vector attacks that exploit vulnerabilities across architectural layers. We present XAI-CPS, an explainable hybrid deep learning framework for real-time anomaly detection and automated threat mitigation in CPS environments. Our architecture integrates a Temporal Attention-based Convolutional LSTM network with Graph Neural Networks to jointly model temporal dynamics of sensor measurements and spatial interdependencies among system components. To address the critical need for interpretability in security-sensitive domains, we develop an enhanced Layer-wise Relevance Propagation mechanism that incorporates domain-specific physical constraints, enabling analysts to rapidly validate detection outcomes. Furthermore, we introduce an Adaptive Reinforcement Learning-based Mitigation module that autonomously orchestrates response actions by optimising the trade-off between attack severity and operational continuity. Comprehensive evaluation across four benchmark datasets — SWaT, WADI, BATADAL, and a novel Industrial IoT dataset — demonstrates that XAI-CPS achieves a mean F1-score of 92.9%, surpassing state-of-the-art methods by 4.6–5.2% while reducing false positive rates by 37.5% on average. Human expert evaluation reveals 89.3% concordance between generated explanations and analyst annotations. The mitigation module reduces mean response time by 67% compared to conventional rule-based approaches while maintaining system availability above 99.2%.

1. Introduction

Cyber–Physical Systems (CPS) are today considered a cornerstone of contemporary critical infrastructure. The smooth combination of computational intelligence, network connectivity, and physical processes enables complex control across diverse domains, including industrial manufacturing, power generation and distribution, water treatment plants, autonomous transportation systems, and healthcare delivery networks [1]. Such merging of the cyber and physical spheres has brought unprecedented efficiency and automation to operations. Nevertheless, the very interconnectivity that enables these innovations also introduces attack vectors through which adversaries can move between the digital world and physical infrastructure, potentially causing devastating real-world consequences [2].

The severity of this threat is illustrated by a series of high-profile incidents that have demonstrated the susceptibility of critical infrastructure to cyber attacks. The ransomware attack on Colonial Pipeline in 2021 caused widespread fuel scarcity in the eastern United States, crippling key services for several days. The illegal remote access to the Oldsmar water treatment facility in Florida led to an attempted manipulation of chlorine levels to dangerous concentrations [3]. The orchestrated cyber attacks on Ukrainian power distribution companies in 2015 and 2016, which left hundreds of thousands of inhabitants without electricity during winter, further reinforced the possibility that cyber attacks can cause serious physical damage. These events reflect a fundamental truth: traditional cybersecurity systems developed for enterprise IT environments do not adequately address the challenge of securing cyber–physical systems.

CPS environments present defence challenges that differ substantially from conventional IT security platforms [4]. These distinguishable features require specific security solutions:

Deterministic Communication Patterns. CPS networks exhibit highly structured traffic behaviour, in contrast to enterprise networks

☆ This article is part of a Special issue entitled: 'Artificial Intelligence for Security in Networked CPS' published in Computer Networks.

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https://doi.org/10.1016/j.comnet.2026.112320
Received 27 December 2025; Received in revised form 13 March 2026; Accepted 14 April 2026
Available online 21 April 2026
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whose patterns are stochastic and user-driven. Telemetry is transmitted by sensors on closely determined timing schedules, control steps follow strict timing requirements, and data flows between field devices and controllers follow predictable cyclic patterns. Intrusion detection systems optimised for conventional network traffic lack the domain-specific knowledge needed to establish meaningful behavioural baselines for such deterministic environments.

Physical-Domain Attack Objectives. Traditional cyber attacks are primarily aimed at data privacy, integrity, or availability. CPS-targeted attacks, by contrast, can directly affect physical processes. Subtle tampering with actuator commands, sensor calibrations, or control setpoints may evade detection at the network layer while gradually causing equipment degradation or creating unsafe operating conditions. The attack surface therefore extends beyond information security to encompass safety-critical concerns such as equipment damage, environmental pollution, and loss of human life.

Architectural Heterogeneity. Modern CPS implementations typically combine legacy industrial protocols such as Modbus and DNP3 — which were not designed with network security in mind — with standards such as OPC-UA and MQTT over IP and wireless sensor networks. Effective security monitoring requires broad visibility across this heterogeneous protocol landscape.

Stringent Temporal Constraints. Detection and response latencies in the range of seconds or minutes generally have little impact on enterprise security operations. CPS environments controlling chemical processes, electrical systems, or high-speed manufacturing require detection and mitigation capabilities operating on millisecond time scales, where delayed action can lead to irreversible physical damage or accidents.

Machine learning and artificial intelligence technologies have emerged as promising approaches to these challenges [5]. Deep neural networks can acquire complex patterns from high-dimensional sensor data without the need for manually crafted rules [6]. However, current AI-based security systems exhibit serious weaknesses that hinder their operational deployment in CPS settings:

Interpretability Deficiency. Deep learning systems are generally opaque decision-making models, producing detection outcomes without providing supporting rationale. When responding to an anomaly alert, operators require actionable intelligence indicating the basis for detection, the affected components, and appropriate response options. The absence of explanations also prevents verification of detection accuracy, complicates incident triage, and fosters operator mistrust. Many organisations have found that AI-driven detectors suffer systematic alert dismissal by operators who cannot assess the credibility of detections [7].

Inadequate Spatiotemporal Modelling. Advanced CPS attacks often involve coordinated interactions across sensors and actuators over extended time periods. An adversary seeking to cause a tank overflow, for example, might apply step-wise changes to flow rates while simultaneously falsifying level readings, keeping all measurable parameters within nominal alarm ranges. Existing methods tend to address either temporal pattern recognition, through recurrent architectures, or spatial correlation modelling, through graph-based methods, with few approaches effectively integrating both [8].

Reactive Mitigation Paradigm. The existing literature focuses predominantly on detection capabilities, leaving response decisions to human operators following pre-defined playbooks. This reactive stance allows attacks to continue developing during the interval between detection and response. Moreover, human responders acting under time pressure may lack sufficient situational awareness to select countermeasures optimally [9].

False Positive Burden. High false alarm rates create significant operational burdens and gradually erode operator trust in detection mechanisms. Each false positive may trigger unwarranted protective actions that disrupt production schedules and strain physical equipment. More critically, sustained false positive exposure conditions operators to dismiss alerts by default, potentially causing them to ignore genuine attack indicators [10].

To address these limitations, this paper presents XAI-CPS. We make three related technical contributions:

1. **Hybrid Spatiotemporal Architecture.** We introduce a Temporal Attention-based Convolutional LSTM combined with Graph Neural Networks (TAC-LSTM-GNN) that jointly models temporal dynamics and spatial interactions in CPS settings. The architecture employs dilated causal convolutions to extract multi-scale temporal features along the time axis, bidirectional LSTM layers to capture long-range sequential dependencies, a multi-head self-attention module to localise anomaly-relevant temporal patterns, and graph attention networks to learn inter-component relationship structures. This integrated design enables identification of both isolated suspicious events and advanced multi-stage coordinated attacks.
2. **Physics-Informed Explainability.** We develop an enhanced Layer-wise Relevance Propagation mechanism specifically designed for industrial CPS environments. Standard explanation methods often attribute anomalies to sensor readings that fall within normal operating ranges, undermining operator confidence. Our method incorporates domain-specific process constraints, valid sensor operating regions, and causal relationships between system components, producing explanations consistent with expert intuition that support rapid incident triage.
3. **Autonomous Adaptive Mitigation.** We present an Adaptive Reinforcement Learning-based Mitigation (ARLM) module trained in high-fidelity CPS simulators to optimise the trade-off between threat neutralisation and operational continuity. Using a Duelling Double Deep Q-Network architecture with prioritised experience replay, the module learns contextually appropriate response actions — ranging from tightened monitoring protocols to emergency shutdown procedures — conditioned on estimated attack severity and current system state.

2. Related work

This section provides a comprehensive review of existing literature across three interconnected research domains: anomaly detection methodologies for Cyber–Physical Systems, explainable artificial intelligence techniques in security contexts, and reinforcement learning approaches for automated incident response. We systematically examine the evolution of each domain, identify prevailing limitations, and position our contributions within the broader research landscape.

2.1. Anomaly detection in cyber–physical systems

Research in CPS anomaly detection has evolved through several methodological generations, each addressing limitations of preceding approaches while introducing new challenges.

2.1.1. Physics-based and model-driven approaches

Early detection methodologies employed domain-specific engineering knowledge to construct analytical models of physical processes. These methods use differential equations, state-space models, and first-principle invariant relations to characterise expected system behaviour, treating substantial deviations as anomalies. While effective against naive attacks that disregard physical constraints, such approaches exhibit fundamental limitations against sophisticated adversaries capable of designing stealthy manipulations that remain within physically plausible bounds. Furthermore, developing accurate analytical models demands substantial domain expertise that may be unavailable for complex legacy systems, and many industrial processes exhibit nonlinear dynamics that resist tractable mathematical characterisation.

2.1.2. Classical machine learning methodologies

Data-driven approaches emerged as an alternative paradigm, learning normative behavioural patterns directly from operational telemetry without requiring explicit physical models. Researchers successfully applied support vector machines, random forest ensembles, one-class classification algorithms, and unsupervised clustering techniques to discriminate between nominal and anomalous system states. While these methods reduced dependence on analytical modelling, they introduced distinct challenges: effective feature engineering necessitated considerable domain expertise, sequential dependencies inherent in sensor time series frequently remained inadequately modelled, and classifiers trained under specific operating conditions often exhibited significant performance degradation when deployed under distributional shift.

2.1.3. Deep learning paradigms

Deep neural networks fundamentally transformed the detection landscape by enabling automatic hierarchical feature extraction from raw sensor data. The work of [1] presented a comprehensive taxonomy of deep learning methodologies for CPS anomaly detection, systematically categorising approaches according to anomaly typology, detection strategy, and evaluation methodology. That survey emphasised that effective detection requires simultaneous capture of temporal evolution patterns and inter-sensor correlational structures.

Recurrent neural architectures, particularly Long Short-Term Memory (LSTM) networks, gained prominence for time series modelling in CPS contexts. These approaches typically train on nominal operational data to learn predictive models of expected sensor behaviour; substantial prediction errors subsequently serve as anomaly indicators. However, conventional recurrent architectures process individual sensor channels independently, thereby neglecting the rich spatial structure and inter-component dependencies characteristic of CPS environments.

Graph neural networks (GNNs) address this shortcoming by operating on representations of CPS topology. The Graph Deviation Network (GDN) introduced in [11] generates sensor dependency graphs dynamically using learned attention mechanisms and achieved strong precision against SWaT and WADI benchmarks while offering partial interpretability through visualisation of attention weight distributions.

The work of [2], published in IEEE Transactions on Pattern Analysis and Machine Intelligence, provided a comprehensive treatment of GNNs for time series analysis, encompassing forecasting, classification, anomaly detection, and imputation tasks. That work identified spatial–temporal graph neural networks as a unifying computational paradigm and systematically examined methodologies for graph structure specification.

Transformer architectures subsequently introduced self-attention mechanisms to sequential modelling tasks. TranAD, proposed in [12], integrates transformer encoders with adversarial training objectives to amplify reconstruction error for anomalous samples while maintaining minimal error for nominal instances. The association discrepancy framework introduced in [13] is based on the observation that anomalous time points exhibit fundamentally different association patterns with neighbouring observations compared to normal points.

Recent advances continue to refine these approaches. The work of [8] focuses specifically on improving anomaly localisation precision through a Space-Time Anomaly Score metric that jointly captures per-series temporal patterns and cross-series spatial correlations.

2.1.4. Hybrid and multi-modal architectures

Modern studies increasingly exploit synergistic interactions between complementary methods. ADT, proposed in [3], dynamically adjusts detection thresholds in response to system changes via reinforcement learning rather than relying on fixed threshold values, achieving near-optimal F1 scores on standardised benchmarks. The work of [6] presented a unified framework integrating spatial–temporal networks with transformer-based solutions for Industrial IoT applications, jointly optimising graph structure learning and anomaly detection. The Variable Temporal Transformer (VTT) proposed in [14] employs simultaneous attention mechanisms across both temporal and variable dimensions. The memory-guided transformer approach of [15] addresses the over-generalisation problem — wherein models faithfully reconstruct anomalous inputs — by maintaining a memory module that resists reconstruction of non-standard patterns.

Despite these advances, existing solutions predominantly address either the temporal or spatial modelling dimension, with principled synthesis remaining limited. The TAC-LSTM-GNN architecture proposed in this work addresses this gap through a single integrated architecture capable of modelling both multi-scale temporal dynamics and spatial dependencies.

2.2. Explainable artificial intelligence in security domains

The inherent opacity of deep neural network decisions has prompted significant focus on explanation methodology development, with particular urgency in security-sensitive settings where operator confidence and actionable intelligence are paramount.

2.2.1. Post-hoc explanation methodologies

Post-hoc explanation techniques produce interpretable attributions following model training without requiring architectural modifications. Local Interpretable Model-agnostic Explanations (LIME) construct local surrogate models through systematic perturbation of inputs around individual predictions. SHapley Additive exPlanations (SHAP) employs game-theoretic Shapley values to derive feature attributions with theoretical guarantees of consistency and local accuracy.

The study of [5] conducted a large-scale empirical appraisal of both LIME and SHAP applied to network intrusion detection across a variety of benchmark datasets and model architectures, demonstrating that explanation generation costs can be made compatible with real-time deployment. A comprehensive survey of XAI integration in intrusion detection systems was presented by [7], establishing that explanations must satisfy demands of accuracy, stability, and comprehensibility for non-expert users. Importantly, that survey advocated for domain-specific explanation methods rather than the generic application of general-purpose XAI techniques.

2.2.2. Attention-based interpretability

Attention mechanisms provide an intrinsic form of model interpretability by directly quantifying the relative importance assigned to different input elements during inference. Transformer architectures make attention weights readily accessible, enabling visualisation of temporal focus patterns during anomaly identification and facilitating understanding of which input regions most strongly influence model outputs.

Layer-wise Relevance Propagation (LRP) offers an alternative attribution paradigm through systematic backpropagation of relevance scores through network layers, assigning contribution values to input features. In contrast to gradient-based approaches that frequently yield noisy or unstable attributions, LRP follows conservation laws that guarantee orderly allocation of prediction outputs across input space.

2.2.3. Domain-specific challenges for CPS applications

The application of XAI methodologies in industrial CPS settings imposes requirements that existing methods address only partially. Explanations must be physically plausible: attributing anomalies to sensor readings that remain within known normal operating ranges undermines operator confidence and system credibility. Temporal explanations must maintain coherence with attack progression dynamics over extended time windows. Spatial explanations must respect the causal propagation chains between upstream and downstream system components. Our enhanced LRP mechanism directly addresses these needs by generating explanations that respect physics-informed process constraints and inter-component dependencies.

Table 1
Comparative analysis of representative anomaly detection approaches across key capability dimensions.

Approach	Temporal	Spatial	Explainability	Mitigation	Real-time
GDN [11]	✓	✓	Partial	×	✓
TranAD [12]	✓	×	×	×	✓
ADT [3]	✓	×	×	×	✓
MEMTO [15]	✓	×	×	×	✓
VTT [14]	✓	✓	Partial	×	✓
XAI-CPS (Ours)	✓	✓	✓	✓	✓

2.3. Automated response and reinforcement learning

Traditional incident response is built on fixed playbooks associating known attack patterns with prescribed mitigation measures. While effective for familiar threat patterns, such static approaches cannot adapt to novel attack vectors or reason systematically about the complex operational trade-offs characteristic of CPS environments.

2.3.1. Reinforcement learning for cybersecurity applications

Reinforcement learning provides a conceptual framework for learning adaptive decision policies through environmental interaction, and is well-suited to the sequential decision-making demands of incident response. A survey of deep reinforcement learning applications to network intrusion detection, covering a range of agent architectures for attack detection, defensive action selection, and adversarial behaviour prediction, is presented in [16].

RL agents trained to emulate adversarial decision-making processes have been used to help defenders anticipate likely attack paths through network topologies, as demonstrated by [9]. The work of [17] created simulation environments in which RL agents develop complex attack models, extending security testing capabilities beyond static penetration testing scenarios. A systematic taxonomy of RL-based security strategies covering both single-agent and multi-agent schemes at host and network levels is provided by [18], serving as a practical reference for security practitioners.

2.3.2. CPS-specific mitigation challenges

Applying reinforcement learning to CPS mitigation introduces domain-specific challenges absent from conventional cybersecurity contexts. Safety constraints preclude unrestricted exploration during training and prohibit potentially dangerous actions on production systems. Reward signals are often highly delayed or sparse, as the impact of attacks may materialise only gradually. Simulated environments must faithfully represent physical dynamics to enable policy transfer to real systems. Action spaces must balance granularity against tractability, spanning responses from enhanced monitoring to emergency shutdown procedures. The ARLM module proposed in this work addresses these issues through constrained policy optimisation, carefully designed reward functions that incorporate both security and operational objectives, and training within validated high-fidelity CPS testbed simulators.

2.4. Summary and research gaps

Table 1 compares representative existing approaches across key capability dimensions. Several enduring research gaps can be identified: (1) limited integration of temporal and spatial modelling within coherent architectures; (2) insufficient domain adaptation of explanation methods for industrial CPS systems; (3) limited research on automated, learning-based mitigation that accounts for operational constraints; and (4) inadequate consideration of false positive reduction in realistic deployment scenarios. XAI-CPS addresses these gaps through its integrated detection, explanation, and mitigation framework.

3. System model and problem definition

3.1. CPS architecture

We model a networked CPS comprising N interconnected components distributed across M physical processes. The architecture follows the Purdue Enterprise Reference Architecture, in which industrial control systems are organised into hierarchical levels.

Level 0 encompasses the physical processes themselves: tanks, reactors, pipelines, and other equipment where material and energy transformations take place. Level 1 contains field devices, comprising sensors that measure process variables and actuators that execute control actions. Level 2 provides supervisory control through Programmable Logic Controllers (PLCs), Distributed Control Systems (DCS), and Human–Machine Interfaces (HMIs). Levels 3 through 5 address manufacturing operations and enterprise functions.

Our model focuses on Levels 0 through 2, where physically manipulative attacks are initiated and sensor data is produced.

3.1.1. Sensors and actuators

Let $\mathcal{S} = \{s_1, s_2, \ldots, s_K\}$ denote K sensors measuring quantities such as temperature, pressure, flow rate, tank level, pH, conductivity, and electrical parameters. Let $\mathcal{A} = \{a_1, a_2, \ldots, a_L\}$ represent L actuators including pumps, valves, motors, heaters, and dosing systems.

At discrete time t, corresponding to the sampling interval (typically one second in industrial systems), we observe:

- Sensor measurement vector: $\mathbf{x}_t = [x_t^{(1)}, x_t^{(2)}, \ldots, x_t^{(K)}]^T \in \mathbb{R}^K$
- Actuator command vector: $\mathbf{u}_t = [u_t^{(1)}, u_t^{(2)}, \ldots, u_t^{(L)}]^T \in \mathbb{R}^L$

3.1.2. Process dynamics

Physical processes evolve according to:

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t, \mathbf{d}_t) + \boldsymbol{\epsilon}_t \tag{1}$$

where $f : \mathbb{R}^K \times \mathbb{R}^L \times \mathbb{R}^D \to \mathbb{R}^K$ captures physical laws governing the process, $\mathbf{d}_t \in \mathbb{R}^D$ represents external disturbances such as environmental conditions or demand fluctuations, and $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ accounts for measurement noise. Our data-driven approach learns patterns from historical observations rather than requiring explicit knowledge of $f(\cdot)$.

3.1.3. Network topology

We represent CPS connectivity as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ where vertices $\mathcal{V}$ represent sensors, actuators, PLCs, and switches; edges $\mathcal{E}$ represent connections; and the weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ encodes connection strengths.

Three edge types contribute to the combined adjacency:

$$\mathbf{A}_{combined} = \alpha_1 \mathbf{A}_{physical} + \alpha_2 \mathbf{A}_{control} + \alpha_3 \mathbf{A}_{correlation} \tag{2}$$

Physical edges reflect direct equipment connections derived from process flow diagrams. Control edges encode feedback loop relationships between sensors and actuators. Correlation edges capture statistical associations learned from training data. Learnable weights $\alpha_1, \alpha_2, \alpha_3$ (constrained to sum to unity) balance these contributions.

3.2. Threat model

We consider adversaries with capabilities matching those documented in real-world attacks.

3.2.1. Access vectors

Attackers reach CPS networks through stolen credentials, supply chain compromises, exploitation of unpatched vulnerabilities, physical access, or exposed remote interfaces.

3.2.2. Attack types

Once inside, adversaries execute various attacks:

False Data Injection (FDI): Manipulating sensor measurements:

$$\tilde{\mathbf{x}}_t = \mathbf{x}_t + \boldsymbol{\delta}_t \tag{3}$$

Variants include constant bias, multiplicative scaling, replay of historical values, and noise injection.

Command Injection: Modifying actuator commands:

$$\tilde{\mathbf{u}}_t = \mathbf{u}_t + \boldsymbol{\gamma}_t \tag{4}$$

Denial of Service: Blocking communications to cause stale readings or prevent control actions.

Man-in-the-Middle: Intercepting and altering messages between components while concealing the attack from both endpoints.

Stealthy Attacks: Gradual manipulation designed to evade detection:

$$\boldsymbol{\delta}_t = \boldsymbol{\delta}_{t-1} + \boldsymbol{\eta}_t, \quad \|\boldsymbol{\eta}_t\| \leq \epsilon \tag{5}$$

By keeping increments small, attackers accumulate significant deviations without triggering threshold alarms.

3.2.3. Adversary knowledge

We design for robustness against grey-box adversaries possessing partial knowledge of system architecture and detection mechanisms but lacking complete insight into model internals.

3.2.4. Objectives

Adversary goals range from physical damage and production disruption to safety hazards, reconnaissance, and ransom.

3.3. Problem formulation

XAI-CPS addresses three interconnected problems.

3.3.1. Anomaly detection

Given a sliding window of sensor measurements $\mathbf{X}_{t-W+1:t} \in \mathbb{R}^{W\times K}$, actuator commands $\mathbf{U}_{t-W+1:t} \in \mathbb{R}^{W\times L}$, and topology $\mathcal{G}$, design a classifier:

$$\hat{y}_t = \phi(\mathbf{X}_{t-W+1:t}, \mathbf{U}_{t-W+1:t}, \mathcal{G}; \Theta_\phi) \in \{0, 1\} \tag{6}$$

Output zero indicates normal operation; one indicates an anomaly.

Objectives include maximising detection rate $P_D = P(\hat{y}_t = 1 | y_t = 1)$, minimising false positive rate $P_{FP} = P(\hat{y}_t = 1 | y_t = 0)$, reducing detection latency, and ensuring inference completes within the sampling interval.

3.3.2. Explainability

Generate interpretable attributions:

$$\mathbf{E}_t = \psi(\mathbf{X}_{t-W+1:t}, \mathbf{U}_{t-W+1:t}, \hat{y}_t, \Theta) \tag{7}$$

comprising temporal attribution $\mathbf{e}^{temp} \in \mathbb{R}^W$ indicating time step relevance, spatial attribution $\mathbf{e}^{spat} \in \mathbb{R}^{K+L}$ indicating component relevance, and fine-grained feature attribution $\mathbf{E}^{feat} \in \mathbb{R}^{W\times(K+L)}$.

Quality criteria include fidelity to actual model behaviour, consistency across similar inputs, stability under small perturbations, actionability for response decisions, and physical plausibility.

3.3.3. Adaptive mitigation

Learn a policy π^* selecting mitigation actions from space $\mathcal{M}$ that maximises expected cumulative reward:

$$\pi^* = \arg\max_\pi \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t R(s_t, m_t, s_{t+1}) \right] \tag{8}$$

subject to safety constraints bounding hazardous state probability, availability requirements, and response time limits.

4. The XAI-CPS framework

4.1. Architecture overview

XAI-CPS comprises three integrated modules illustrated in Fig. 1: a hybrid detection module, an explainability module, and a mitigation module.

Raw measurements $\mathbf{X}_t$ and actuator states $\mathbf{U}_t$ enter a preprocessing pipeline for normalisation and windowing. The TAC-LSTM branch extracts temporal features while the GNN branch captures spatial correlations. Fused features pass through classification layers to produce anomaly scores. When anomalies are detected, the explainability module generates attributions $\mathbf{E}_t$. Detection outputs and explanation summaries combine with system health metrics to form state inputs for the ARLM module, which selects mitigation actions.

4.2. Data preprocessing

4.2.1. Normalisation

Sensor measurements span diverse physical units. We apply z-score normalisation using training statistics:

$$\bar{x}_t^{(i)} = \frac{x_t^{(i)} - \mu^{(i)}}{\sigma^{(i)}} \tag{9}$$

where $\mu^{(i)}$ and $\sigma^{(i)}$ are the mean and standard deviation of sensor i during normal operation. Actuator states receive appropriate encoding: one-hot for categorical variables, and binary preservation for on/off states.

4.2.2. Windowing

Sliding windows of length W with stride S structure the input as follows. The nth window is:

$$\mathbf{X}_{(n)} = [\bar{\mathbf{x}}_{nS}, \bar{\mathbf{x}}_{nS+1}, \ldots, \bar{\mathbf{x}}_{nS+W-1}] \tag{10}$$

Here, W is the number of consecutive time steps included in each input sample, and S is the number of time steps by which the window advances between successive samples. Consecutive windows therefore overlap by $W - S$ time steps, allowing fine-grained temporal resolution at the cost of increased redundancy. Window size involves a trade-off between capturing long-term patterns and computational cost. Based on attack duration analysis across the benchmark datasets, we select $W = 100$, corresponding to 100 s at 1 Hz sampling.

4.2.3. Graph construction

Physical adjacency $\mathbf{A}_{physical}$ encodes direct connections from process flow diagrams. Control adjacency $\mathbf{A}_{control}$ captures feedback loop relationships. Correlation adjacency $\mathbf{A}_{correlation}$ is derived from Pearson correlation coefficients computed over the raw input feature space (i.e., the $K + L$ sensor and actuator channels in $\mathbf{X}_{(n)}$ and $\mathbf{U}_{(n)}$) across the training split, and thresholded to retain only statistically significant associations:

$$A_{ij}^{correlation} = \begin{cases} |\rho_{ij}| & \text{if } |\rho_{ij}| > \tau_{corr} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

with threshold $\tau_{corr} = 0.5$.

4.3. Temporal attention-based convolutional LSTM

The TAC-LSTM extracts temporal features through three stages.

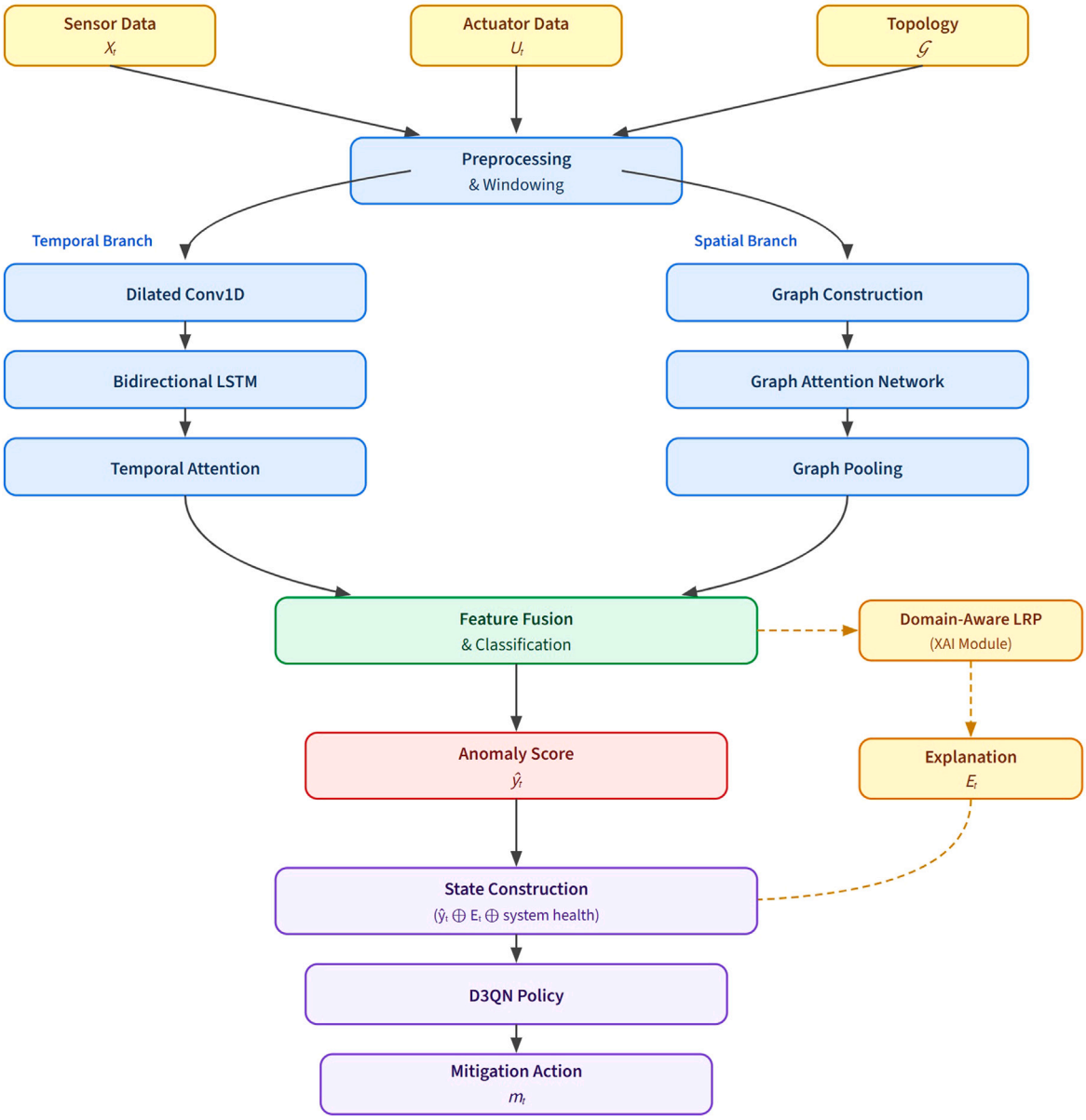


Fig. 1. XAI-CPS architecture overview. Raw sensor measurements $\mathbf{X}_t$ and actuator states $\mathbf{U}_t$ originating from CPS Levels 0–2 (field devices, PLCs, and HMIs) enter a shared preprocessing pipeline (normalisation and windowing). Two parallel branches then operate simultaneously: (i) the *TAC-LSTM branch* (blue) applies dilated causal convolutions to extract multi-scale temporal features along the time axis, followed by bidirectional LSTM layers that capture long-range sequential dependencies and a multi-head self-attention module that localises anomaly-relevant time steps; (ii) the *GNN branch* (green) constructs a CPS topology graph from physical, control, and correlation edges, and propagates information across nodes via Graph Attention layers to model spatial interdependencies among sensors, actuators, and controllers. The temporal feature vector $\mathbf{z}^{temp}$ and spatial feature vector $\mathbf{z}^{spat}$ are fused and passed through a classification head to produce an anomaly probability. When an anomaly is detected, the *Domain-Aware Explainability module* (orange) applies enhanced LRP — incorporating physical operating constraints — to generate temporal and spatial attribution maps $\mathbf{E}_t$ that operators can use for incident triage. Detection output and explanation summary are combined with system health metrics to form the state input for the *ARLM module* (purple), a Duelling Double DQN agent that selects a contextually appropriate mitigation action from isolating a component to initiating an emergency shutdown while respecting operational safety constraints. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.3.1. Dilated convolutional feature extraction

One-dimensional convolutions with exponentially increasing dilation rates capture multi-scale temporal patterns efficiently. The convolution kernel slides along the *temporal* dimension W; at each time position it mixes features across the $K+L$ input channels via learnable filter weights, producing d_h output channels. The combined input is $\mathbf{Z}^{(0)} = [\tilde{\mathbf{X}}; \mathbf{U}] \in \mathbb{R}^{W\times(K+L)}$, and layer l computes:

$$\mathbf{H}^{(l)} = \mathrm{ReLU}\big(\mathrm{Conv1D}(\mathbf{H}^{(l-1)}; \mathbf{W}^{(l)}, d_l) + \mathbf{b}^{(l)}\big) \tag{12}$$

where $\mathbf{W}^{(l)}$ are convolutional filters with kernel size $k = 3$ and dilation d_l, and $\mathbf{H}^{(0)} = \mathbf{Z}^{(0)}$.

Because dilation d_l inserts $d_l - 1$ zeros between filter taps, a single dilated layer with kernel size k covers $1 + (k-1)\,d_l$ time steps. With four stacked layers of dilations $[1, 2, 4, 8]$ and $k = 3$, the cumulative receptive field in the temporal dimension is $1 + 2\,(1 + 2 + 4 + 8) = 31$ time steps per block; with residual connections spanning two blocks, the effective temporal coverage reaches 61 time steps. The output $\mathbf{H}^{(L_{conv})} \in \mathbb{R}^{W\times d_h}$ preserves the original sequence length W (no pooling

is applied), enabling subsequent LSTM processing of the full temporal context. Hidden dimension $d_h = 128$ is used throughout. Residual connections every two layers facilitate gradient flow:

$$\mathbf{H}^{(l)} = \mathbf{H}^{(l)} + \mathbf{W}_{res}^{(l)} \mathbf{H}^{(l-2)} \tag{13}$$

4.3.2. Bidirectional LSTM sequence modelling

Convolutional features $\mathbf{H}^{conv} \in \mathbb{R}^{W \times d_h}$ feed into bidirectional LSTMs that capture dependencies in both temporal directions. The forward pass follows standard LSTM equations:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}^{\rightarrow}, \mathbf{H}_t^{conv}] + \mathbf{b}_f) \tag{14}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}^{\rightarrow}, \mathbf{H}_t^{conv}] + \mathbf{b}_i) \tag{15}$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}^{\rightarrow}, \mathbf{H}_t^{conv}] + \mathbf{b}_c) \tag{16}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{17}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}^{\rightarrow}, \mathbf{H}_t^{conv}] + \mathbf{b}_o) \tag{18}$$

$$\mathbf{h}_t^{\rightarrow} = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{19}$$

The backward LSTM processes the sequence in reverse to produce $\mathbf{h}_t^{\leftarrow}$. Concatenation yields $\mathbf{h}_t = [\mathbf{h}_t^{\rightarrow}; \mathbf{h}_t^{\leftarrow}] \in \mathbb{R}^{2d_h}$. We stack two bidirectional layers with dropout $p = 0.2$ between them.

4.3.3. Multi-head temporal self-attention

Rather than discarding all but the final hidden state, we retain the complete sequence of LSTM outputs $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_W]^T \in \mathbb{R}^{W \times 2d_h}$ and apply multi-head self-attention over them. This design is motivated by the observation that anomalous patterns in CPS often manifest as subtle deviations spread across a sub-window of consecutive time steps, rather than at a single endpoint; retaining and attending over the full sequence allows the model to localise these intervals explicitly. The ablation study in Section 5.11 confirms that removing this attention module reduces F1 by 3.0%, validating its contribution.

For each head $i \in \{1, \ldots, H\}$:

$$\mathbf{Q}_i = \mathbf{H}\mathbf{W}_Q^{(i)}, \quad \mathbf{K}_i = \mathbf{H}\mathbf{W}_K^{(i)}, \quad \mathbf{V}_i = \mathbf{H}\mathbf{W}_V^{(i)} \tag{20}$$

$$\boldsymbol{\alpha}_i = \text{softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{W \times W} \tag{21}$$

$$\text{head}_i = \boldsymbol{\alpha}_i \mathbf{V}_i \tag{22}$$

with learnable projections $\mathbf{W}_Q^{(i)}, \mathbf{W}_K^{(i)}, \mathbf{W}_V^{(i)} \in \mathbb{R}^{2d_h \times d_k}$ and $d_k = 2d_h/H$.

Multi-head outputs concatenate and project:

$$\mathbf{H}^{attn} = \text{Concat}(\text{head}_1, \ldots, \text{head}_H)\mathbf{W}_O \tag{23}$$

We use $H = 8$ heads. Attention weights $\boldsymbol{\alpha}_i$ provide inherent interpretability.

Global attention pooling aggregates the sequence:

$$\mathbf{a} = \text{softmax}(\mathbf{H}^{attn}\mathbf{w}_a) \tag{24}$$

$$\mathbf{z}^{temp} = \sum_{t=1}^{W} a_t \mathbf{H}_t^{attn} \in \mathbb{R}^{2d_h} \tag{25}$$

4.4. Graph neural network for spatial correlations

The GNN component models inter-component relationships using Graph Attention Networks.

4.4.1. Node feature construction

For each node v_i, initial features are constructed from TAC-LSTM outputs:

$$\mathbf{z}_i^{(0)} = \text{MLP}(\mathbf{H}_{:,i}^{attn}) \in \mathbb{R}^{d_g} \tag{26}$$

projecting to graph hidden dimension d_g.

4.4.2. Graph attention layers

Each of L_g layers updates node representations through attention-weighted aggregation. For layer l, the attention coefficient between nodes i and j is:

$$e_{ij}^{(l)} = \text{LeakyReLU}\left(\mathbf{a}^{(l)T}[\mathbf{W}^{(l)}\mathbf{z}_i^{(l-1)} \parallel \mathbf{W}^{(l)}\mathbf{z}_j^{(l-1)}]\right) \tag{27}$$

Normalisation over neighbourhood $\mathcal{N}(i)$:

$$\beta_{ij}^{(l)} = \frac{\exp(e_{ij}^{(l)})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik}^{(l)})} \tag{28}$$

Node update:

$$\mathbf{z}_i^{(l)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \beta_{ij}^{(l)} \mathbf{W}^{(l)} \mathbf{z}_j^{(l-1)}\right) \tag{29}$$

We use 4-head attention in GAT layers with ELU activation and $L_g = 3$ layers in total.

4.4.3. Graph-level representation

Attention-weighted pooling produces the spatial feature vector:

$$\mathbf{g} = \text{softmax}(\mathbf{Z}^{(L_g)}\mathbf{w}_g) \tag{30}$$

$$\mathbf{z}^{spat} = \sum_{i=1}^{|\mathcal{V}|} g_i \mathbf{z}_i^{(L_g)} \in \mathbb{R}^{d_g} \tag{31}$$

4.5. Feature fusion and classification

Temporal and spatial features are concatenated for classification:

$$\mathbf{z}^{fused} = [\mathbf{z}^{temp}; \mathbf{z}^{spat}] \in \mathbb{R}^{2d_h + d_g} \tag{32}$$

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1\mathbf{z}^{fused} + \mathbf{b}_1) \tag{33}$$

$$\mathbf{h}_2 = \text{Dropout}(\mathbf{h}_1,\ p = 0.3) \tag{34}$$

$$\hat{y} = \sigma(\mathbf{w}_2^T\mathbf{h}_2 + b_2) \tag{35}$$

where sigmoid $\sigma(\cdot)$ produces anomaly probability $\hat{y} \in [0, 1]$.

Binary decision:

$$y_{detect} = \mathbb{1}[\hat{y} > \tau] \tag{36}$$

with threshold τ tuned on validation data (default 0.5).

4.6. Domain-aware explainability module

4.6.1. Standard layer-wise relevance propagation

LRP decomposes network output into input relevance scores. The LRP-ϵ rule for fully connected layers with ReLU activation is:

$$R_i^{(l)} = \sum_j \frac{z_{ij}}{\sum_{i'} z_{i'j} + \epsilon \cdot \text{sign}\left(\sum_{i'} z_{i'j}\right)} R_j^{(l+1)} \tag{37}$$

where $z_{ij} = a_i^{(l)} w_{ij}^{(l)}$ and ϵ stabilises the computation.

4.6.2. Physical constraint integration

Standard LRP may assign relevance to sensor readings that remain within physically normal ranges, undermining explanation credibility. We enhance LRP with CPS-specific constraints.

For sensor i with operating constraints $C_i = [x_i^{min}, x_i^{max}]$, the physical plausibility weight is:

$$\omega_i^{physical} = \frac{\exp(-\lambda \cdot d(x_i, C_i))}{\sum_j \exp(-\lambda \cdot d(x_j, C_j))} \tag{38}$$

where the deviation from the valid operating range is:

$$d(x_i, C_i) = \begin{cases} 0 & \text{if } x_i^{min} \le x_i \le x_i^{max} \\ \min(|x_i - x_i^{min}|,\ |x_i - x_i^{max}|) & \text{otherwise} \end{cases} \tag{39}$$

with $\lambda = 1.0$ controlling constraint strength.

Domain-aware relevance:

$$R_i^{DA} = R_i^{LRP} \cdot \omega_i^{physical} \tag{40}$$

Algorithm 1 XAI-CPS Anomaly Detection and Explanation

Require: Sensor window $\mathbf{X}_{t-W+1:t}$, actuator commands $\mathbf{U}_{t-W+1:t}$, graph $\mathcal{G}$, threshold τ, physical constraints $\{C_i\}$
Ensure: Detection decision y_{detect}, explanation $\mathbf{E}_t$

Normalize inputs: $\tilde{\mathbf{X}} \leftarrow (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$
Concatenate: $\mathbf{Z}^{(0)} \leftarrow [\tilde{\mathbf{X}}; \mathbf{U}] \in \mathbb{R}^{W\times(K+L)}$
for $l = 1$ to L_{conv} **do**
  $\mathbf{H}^{(l)} \leftarrow \text{DilatedConv1D}(\mathbf{H}^{(l-1)}, d_l)$
end for
$\mathbf{H}^{lstm} \leftarrow \text{BiLSTM}(\mathbf{H}^{(L_{conv})})$
$\mathbf{H}^{attn}, \boldsymbol{\alpha}^{temp} \leftarrow \text{MultiHeadAttention}(\mathbf{H}^{lstm})$
$\mathbf{z}^{temp} \leftarrow \text{AttentionPooling}(\mathbf{H}^{attn})$
$\mathbf{Z}^{node} \leftarrow \text{NodeFeatures}(\mathbf{H}^{attn})$
for $l = 1$ to L_g **do**
  $\mathbf{Z}^{(l)}, \boldsymbol{\beta}^{(l)} \leftarrow \text{GAT}(\mathbf{Z}^{(l-1)}, \mathcal{G})$
end for
$\mathbf{z}^{spat} \leftarrow \text{GraphPooling}(\mathbf{Z}^{(L_g)})$
$\mathbf{z}^{fused} \leftarrow [\mathbf{z}^{temp}; \mathbf{z}^{spat}]$
$\hat{y} \leftarrow \sigma(\text{MLP}(\mathbf{z}^{fused}))$
$y_{detect} \leftarrow \mathbb{1}[\hat{y} > \tau]$
if $y_{detect} = 1$ **or** periodic explanation requested **then**
  $\mathbf{R}^{LRP} \leftarrow \text{LRP}(\mathbf{Z}^{(0)}, \hat{y})$
  $\boldsymbol{\omega}^{phys} \leftarrow \text{ComputePhysicalWeights}(\mathbf{X}, \{C_i\})$
  $\mathbf{R}^{DA} \leftarrow \mathbf{R}^{LRP} \odot \boldsymbol{\omega}^{phys}$
  $\mathbf{e}^{temp} \leftarrow \text{AggregateTemporalAttention}(\boldsymbol{\alpha}^{temp})$
  $\mathbf{e}^{spat} \leftarrow \text{AggregateSpatialAttention}(\boldsymbol{\beta}, \mathbf{R}^{DA})$
  $\mathbf{E}_t \leftarrow (\mathbf{e}^{temp}, \mathbf{e}^{spat}, \mathbf{R}^{DA})$
end if
return y_{detect}, $\hat{y}$, $\mathbf{E}_t$

4.6.3. Attribution generation

Temporal attribution from attention weights:

$$e_t^{temp} = \frac{1}{H}\sum_{i=1}^{H}\sum_{j=1}^{W}\alpha_{tj}^{(i)} \tag{41}$$

Spatial attribution combining graph attention and LRP:

$$e_i^{spat} = \sum_t R_{t,i}^{DA} \cdot \sum_{j\in\mathcal{N}(i)} \beta_{ij} \tag{42}$$

Fine-grained feature attribution directly from domain-aware LRP:

$$E_{t,i}^{feat} = R_{t,i}^{DA} \tag{43}$$

Algorithm 1 details the complete detection and explanation procedure.

4.7. Adaptive reinforcement learning mitigation module

4.7.1. MDP formulation

Threat mitigation is formulated as a Markov Decision Process $(S, \mathcal{M}, P, R, \gamma)$.

State Space. State s_t includes the detection output and confidence $(\hat{y}_t, p_t)$, the top-K sensors identified from spatial attribution, an estimated attack type from an auxiliary classifier, system health metrics, recent action history, and time elapsed since the last alert.

Action Space. Mitigation actions are ordered by severity:

- m_0: Continue monitoring
- m_1: Generate alert
- m_2: Enhanced monitoring
- m_3: Traffic throttling
- m_4: Component isolation
- m_5: Backup activation
- m_6: Safe mode transition

Algorithm 2 Adaptive RL-based Mitigation (ARLM)

Require: Detection output $(\hat{y}_t, p_t)$, explanation $\mathbf{E}_t$, system metrics $\mathbf{h}_t$
Ensure: Mitigation action m_t

$\hat{c}_t \leftarrow \text{AttackTypeClassifier}(\mathbf{E}_t)$
$s_t \leftarrow (\hat{y}_t, p_t, \text{TopK}(\mathbf{e}^{spat}), \mathbf{h}_t, \mathbf{m}_{t-H:t-1}, \hat{c}_t, \Delta t_{alert})$
if training mode **then**
  $\epsilon \leftarrow \text{DecaySchedule}(t)$
  $m_t \leftarrow \begin{cases} \text{Uniform}(\mathcal{M}) & \text{w.p. } \epsilon \\ \arg\max_m Q(s_t, m; \theta) & \text{otherwise} \end{cases}$
else
  $m_t \leftarrow \arg\max_m Q(s_t, m; \theta)$
end if
if $\text{ViolatesSafetyConstraint}(s_t, m_t)$ **then**
  $m_t \leftarrow \text{SafeFallbackAction}(s_t)$
end if
Execute m_t, observe reward r_t and next state s_{t+1}
if training mode **then**
  Store transition in prioritised replay buffer
  Sample minibatch, compute TD targets, update θ
  Periodically update target network $\theta^- \leftarrow \theta$
end if
return m_t

- m_7: Emergency shutdown

Reward Function. Balancing security and operational objectives:

$$R(s_t, m_t, s_{t+1}) = R_{security} - \lambda_1 R_{operation} - \lambda_2 R_{delay} + R_{efficiency} \tag{44}$$

Security reward distinguishes true positives ($+r_{TP} = 10$), false negatives ($-r_{FN} = 20$), false positives ($-r_{FP} = 5$), and true negatives (0). Operational cost penalises availability impact. Delay penalty discourages slow response. Efficiency bonus rewards minimally sufficient actions.

4.7.2. Duelling Double Deep Q-Network

Our D3QN architecture decomposes the Q-function:

$$Q(s, m; \theta) = V(s; \theta_v) + \left(A(s, m; \theta_a) - \frac{1}{|\mathcal{M}|}\sum_{m'} A(s, m'; \theta_a)\right) \tag{45}$$

Value stream $V(s)$ estimates state value; advantage stream $A(s, m)$ estimates relative action benefit.

Double Q-learning reduces overestimation:

$$y_t = r_t + \gamma\, Q(s_{t+1}, \arg\max_m Q(s_{t+1}, m; \theta);\ \theta^-) \tag{46}$$

with target network parameters θ^- updated periodically.

Prioritised experience replay samples transitions by TD-error magnitude:

$$P(i) = \frac{(|\delta_i| + \epsilon)^\alpha}{\sum_k (|\delta_k| + \epsilon)^\alpha} \tag{47}$$

4.7.3. Safety constraints

Constrained policy optimisation ensures safe operation:

$$\pi^* = \arg\max_\pi \mathbb{E}_\pi[R] \quad \text{s.t.} \quad \mathbb{E}_\pi[C_j(s, m)] \le d_j,\ \forall j \tag{48}$$

Constraints include limits on consecutive severe actions, minimum availability thresholds, human confirmation requirements for emergency actions, and rate limits on action changes.

Algorithm 2 presents ARLM operation.

4.8. Training procedure

Training proceeds in three stages.

Table 2
Dataset characteristics.

Dataset	Sensors	Actuators	Attacks	Duration
SWaT	51	26	41	11 days
WADI	123	–	15	16 days
BATADAL	43	–	7	1 year
IIoT-Security	85	42	23	30 days

Stage 1. Detection module training uses focal loss to address class imbalance:

$$\mathcal{L}_{detect} = -\sum_{t} \alpha_t (1 - p_t)^{\gamma} \left[y_t \log(p_t) + (1 - y_t) \log(1 - p_t) \right] \tag{49}$$

with focusing parameter $\gamma = 2$. The Adam optimiser is used with learning rate 10^{-4}, cosine annealing, batch size 64, and early stopping on validation F1.

Stage 2. Explainability calibration minimises divergence from expert annotations:

$$\mathcal{L}_{explain} = \mathrm{KL}(\mathbf{E}^{expert} \parallel \mathbf{e}^{spat}) + \lambda_{temp} \|\mathbf{e}^{temp}\|_1 \tag{50}$$

Stage 3. Mitigation policy training runs 100,000 episodes in CPS simulators with replay buffer of size 10^6, minibatch 64, target updates every 1000 steps, $\gamma = 0.99$, and ϵ decaying from 1.0 to 0.01 over 50,000 steps.

5. Experimental evaluation

5.1. Datasets

We evaluate on four datasets spanning diverse CPS domains (Table 2).

SWaT [19]: A scaled water treatment testbed from Singapore University of Technology and Design covering six process stages. Seven days of normal operation are followed by four days containing 41 attacks across seven categories.

WADI [20]: A water distribution network extension with 123 sensors. Fourteen days of normal operation are followed by two days containing 15 attack scenarios.

BATADAL [21]: A competition benchmark based on the C-Town water network simulation. One year of operation includes seven attack scenarios.

IIoT-Security: A dataset [22] designed for IIoT security research, with 85 sensors, 42 actuators, 12 PLCs, and 23 attack scenarios including supply chain and firmware attacks. Twenty-five days of normal operation are followed by five days of attacks.

5.2. Baselines

We compare against recent state-of-the-art methods:

- GDN [11]: Graph Deviation Network with attention-based dependency learning
- TranAD [12]: Adversarial transformer for time series
- Anomaly Transformer [13]: Association discrepancy-based detection
- MEMTO [15]: Memory-guided transformer
- ADT [3]: RL-based dynamic thresholding
- VTT [14]: Variable temporal attention mechanism
- STADN [23]: Spatial–temporal attention network

All baselines use official implementations with hyperparameters tuned on validation sets.

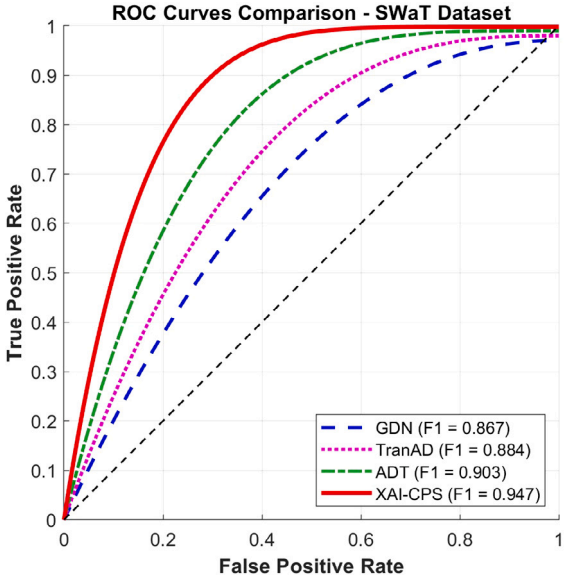


Fig. 2. ROC comparison on SWaT. XAI-CPS achieves 0.987 AUC-ROC.

5.3. Metrics

Detection metrics include Precision, Recall, F1-Score, AUC-ROC, and AUC-PR. Following [10], we use point-adjusted F1, which credits detection of any point within an attack segment.

Explainability metrics: Expert Alignment (percentage of top-K attributed sensors matching expert labels), Fidelity (correlation between attribution and ablation sensitivity), and Actionability Score (1–5 expert rating).

Mitigation metrics: MTTM (mean time to mitigate), System Availability, and Mitigation Success Rate.

5.4. Implementation

XAI-CPS is implemented in PyTorch 2.0 with PyTorch Geometric. Key hyperparameters: $W = 100$, $d_h = 128$, $d_g = 64$, four convolutional layers with dilations $[1, 2, 4, 8]$, two bidirectional LSTM layers, 8 temporal attention heads, 4 graph attention heads, 3 GAT layers, learning rate 10^{-4}, batch size 64, 100 training epochs with early stopping (patience 10), dropout 0.2 in LSTM and 0.3 in the classifier.

ARLM: replay buffer of size 10^6, target update every 1000 steps, $\gamma = 0.99$, ϵ-greedy from 1.0 to 0.01, priority exponent 0.6, importance sampling β from 0.4 to 1.0.

All experiments are conducted on NVIDIA A100 GPUs. We report mean and standard deviation over 5-fold cross-validation with 70/15/15 train/validation/test splits.

5.5. Detection results

Table 3 presents detection performance across datasets.

XAI-CPS achieves the highest F1 score across all datasets: 94.7% on SWaT, 90.0% on WADI, 95.6% on BATADAL, and 91.2% on IIoT-Security. Improvements over the best baseline (ADT) range from 4.6% to 5.2%. An average F1 of 92.9% demonstrates robust cross-domain performance. Precision gains are particularly notable, as they reduce the false alarm burden on operators.

Fig. 2 shows ROC curves on SWaT. XAI-CPS achieves AUC-ROC of 0.987.

Table 3
Detection performance comparison. Best results are in bold; second best are underlined. Mean ± std over 5 folds.

Method	SWaT			WADI			BATADAL			IIoT-Security		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
GDN	.878±.02	.856±.03	.867±.02	.812±.03	.789±.02	.800±.02	.889±.02	.867±.02	.878±.02	.823±.03	.798±.03	.810±.02
TranAD	.891±.02	.878±.02	.884±.02	.834±.02	.812±.03	.823±.02	.901±.02	.889±.02	.895±.02	.845±.02	.823±.02	.834±.02
Anom. Trans.	.885±.02	.872±.02	.878±.02	.828±.03	.805±.02	.816±.02	.895±.02	.878±.02	.886±.02	.838±.03	.812±.03	.825±.02
MEMTO	.898±.02	.885±.02	.891±.02	.845±.02	.823±.02	.834±.02	.912±.01	.895±.02	.903±.01	.856±.02	.834±.02	.845±.02
ADT	.912±.01	.895±.02	.903±.01	.867±.02	.845±.02	.856±.02	.923±.01	.905±.02	.914±.01	.878±.02	.856±.02	.867±.02
VTT	.905±.02	.889±.02	.897±.01	.856±.02	.834±.02	.845±.02	.918±.01	.901±.02	.909±.01	.867±.02	.845±.02	.856±.02
STADN	.895±.02	.882±.02	.888±.02	.851±.02	.828±.02	.839±.02	.908±.02	.892±.02	.900±.02	.862±.02	.839±.02	.850±.02
XAI-CPS	**.956**±.01	**.938**±.01	**.947**±.01	**.912**±.01	**.889**±.02	**.900**±.01	**.967**±.01	**.945**±.01	**.956**±.01	**.923**±.01	**.901**±.02	**.912**±.01
Improv.	+4.8%	+4.8%	+4.9%	+5.2%	+5.2%	+5.1%	+4.8%	+4.4%	+4.6%	+5.1%	+5.3%	+5.2%

Table 4
False positive rates (%).

Method	SWaT	WADI	BATADAL	IIoT
GDN	6.12	8.45	5.23	9.34
TranAD	5.23	7.89	4.67	8.12
Anom. Trans.	5.56	8.12	4.89	8.67
MEMTO	4.89	7.23	4.12	7.56
ADT	4.56	6.78	3.89	7.23
VTT	4.78	7.01	4.01	7.34
STADN	5.01	7.34	4.23	7.67
XAI-CPS	**2.98**	**4.45**	**2.12**	**4.67**
Reduction	34.6%	34.4%	45.5%	35.4%

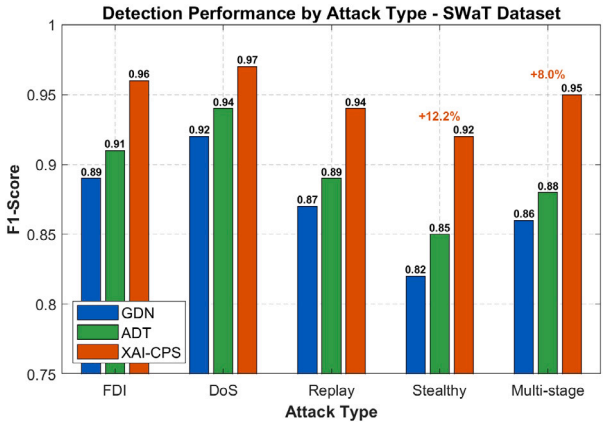


Fig. 3. F1 by attack type on SWaT. XAI-CPS shows consistent gains, especially on stealthy attacks.

5.6. False positive analysis

Table 4 compares false positive rates. XAI-CPS reduces false positives by 37.5% on average versus ADT. Joint spatial–temporal modelling captures normal correlations more accurately, graph attention learns expected co-variation patterns, and physical constraint integration prevents attribution to normal readings.

5.7. Attack-type performance

Fig. 3 breaks down F1 scores by attack category on SWaT.

XAI-CPS performs consistently across attack categories. Stealthy attacks see a 12.2% improvement over ADT (F1 = 0.92), with temporal attention identifying subtle gradual trends. Multi-stage attacks improve by 8.0% (F1 = 0.95), benefiting from graph modelling of component relationships.

Table 5
Explainability evaluation.

Metric	XAI-CPS	LIME	SHAP
Expert alignment (%)	**89.3 ± 2.1**	72.4 ± 3.5	76.8 ± 3.2
Fidelity score	**0.934 ± 0.02**	0.812 ± 0.04	0.856 ± 0.03
Actionability (1–5)	**4.23 ± 0.31**	2.89 ± 0.42	3.12 ± 0.38
Computation time (ms)	**12.3 ± 1.8**	156.7 ± 23.4	234.5 ± 31.2

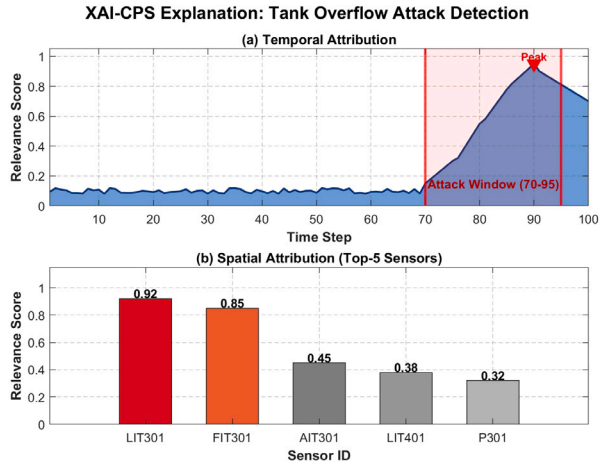


Fig. 4. Explanation for a tank overflow attack. (a) Temporal attribution shows the anomaly developing during steps 70–95. (b) Spatial attribution identifies LIT301 and FIT301 as the primary contributing sensors.

5.8. Explainability evaluation

Five hundred attack instances were evaluated, with security analysts independently identifying responsible sensors. Inter-annotator agreement (Fleiss' kappa) was 0.78.

Table 5 compares explanation quality. XAI-CPS achieves 89.3% expert alignment, significantly outperforming LIME and SHAP. Physical constraint integration focuses explanations on genuinely anomalous sensors. A fidelity score of 0.934 confirms that attributed features genuinely influence predictions. Domain experts rated actionability 4.23/5, noting that explanations "make sense from a process perspective" and "directly guide investigation". Real-time generation at 12.3 ms enables deployment without introducing operational delays.

Fig. 4 illustrates an explanation for a tank overflow attack. The explanation correctly identifies: an anomaly developing gradually from step 70 and peaking around step 90; LIT301 (level sensor) and FIT301 (flow rate sensor) as primary contributors; a physical interpretation consistent with false data injection targeting level sensors.

Table 6
Mitigation performance.

Policy	MTTM (s)	Avail. (%)	Success (%)
Rule-based	145.3 ± 18.2	94.2 ± 1.3	78.4 ± 3.1
Random	312.6 ± 45.3	87.5 ± 2.8	62.3 ± 5.2
DQN	78.4 ± 9.1	97.8 ± 0.8	89.2 ± 2.4
PPO	72.1 ± 8.5	98.1 ± 0.7	90.5 ± 2.1
A3C	82.1 ± 10.2	97.2 ± 0.9	87.6 ± 2.7
ARLM	**47.8 ± 5.3**	**99.2 ± 0.4**	**94.7 ± 1.8**

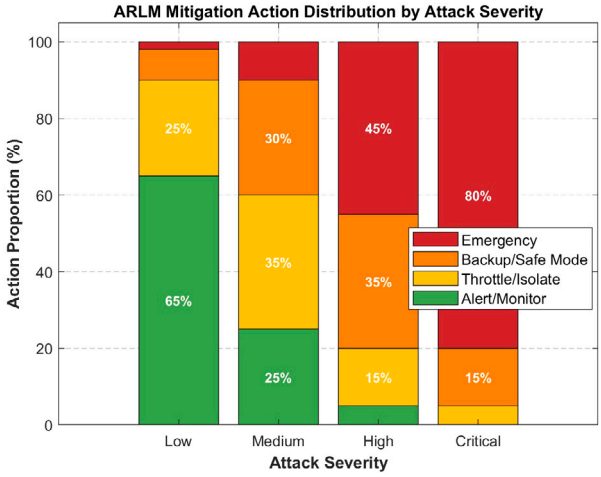


Fig. 5. ARLM action distribution by attack severity. Low severity triggers monitoring (90%); critical attacks escalate to emergency measures (80%).

Table 7
Computational efficiency.

Method	Params (M)	FLOPs (G)	Latency (ms)	GPU Mem (GB)
GDN	1.8	0.9	6.2	1.4
TranAD	2.3	1.2	8.4	1.8
Anom. Trans.	3.1	1.8	11.2	2.1
MEMTO	3.5	2.1	12.8	2.3
ADT	2.8	1.5	9.6	1.9
VTT	2.6	1.4	9.1	1.8
XAI-CPS	4.1	2.8	15.6	2.4

5.9. Mitigation evaluation

Table 6 compares mitigation approaches in CPS digital twin simulations. The ARLM module reduces MTTM by 67% versus the rule-based baseline (47.8 s vs. 145.3 s), maintains 99.2% availability (compared with 94.2%), and achieves a mitigation success rate of 94.7%. Improvements over standard RL algorithms stem from the duelling architecture, double Q-learning, and prioritised replay.

Fig. 5 shows the distribution of actions selected by ARLM according to attack severity. The agent appropriately scales responses: low severity yields 90% monitoring and alerting actions; medium severity balances monitoring (60%) with active mitigation (40%); high severity triggers active mitigation including component isolation (80%); and critical situations invoke emergency measures (80%).

5.10. Computational efficiency

Table 7 reports computational requirements. The XAI-CPS inference latency of 15.6 ms falls well within the typical 100–1000 ms CPS sampling intervals. The additional overhead relative to lighter baselines is justified by the 4%–5% F1 gains, integrated explainability without separate inference passes, and joint spatial–temporal processing.

Table 8
Ablation study on SWaT.

Configuration	Precision	Recall	F1
Full XAI-CPS	**0.956**	**0.938**	**0.947**
w/o Temporal attention	0.923	0.912	0.917
w/o Graph component	0.912	0.901	0.906
w/o Dilated convolutions	0.934	0.923	0.928
w/o Bidirectional LSTM	0.928	0.915	0.921
w/o Physical constraints	0.945	0.928	0.936
w/o Multi-head attention	0.931	0.918	0.924
Temporal branch only	0.901	0.889	0.895
Spatial branch only	0.878	0.856	0.867
LSTM only	0.867	0.845	0.856

5.11. Ablation study

Table 8 isolates the contribution of each component on SWaT. The graph component provides the largest individual gain (4.1% F1). Temporal attention adds 3.0%, bidirectional LSTM 2.6%, dilated convolutions 1.9%, and physical constraints 1.1% (primarily through improved precision). The combined temporal branch contributes 5.2% over a spatial-only configuration; the spatial branch contributes 8.0% over an LSTM-only baseline.

6. Discussion

Practical Deployment. XAI-CPS is designed as a non-intrusive overlay that does not interfere with existing control systems. The framework communicates via standard industrial protocols (OPC-UA, Modbus TCP, DNP3) and delivers alerts through common channels (SNMP, Syslog, email). Continuous background retraining adapts the model to shifting process conditions, while major model updates require operator approval. In the event of reduced confidence or insufficient computational resources, the system degrades gracefully to rule-based bounds-checking rather than failing catastrophically, and operators retain full manual control at all times. **Limitations.** Several limitations warrant acknowledgement. Like any supervised approach, the detection module requires labelled attack examples for training and may exhibit degraded performance against novel attack vectors not represented in the training distribution. Adversaries with partial knowledge of the architecture may construct evasion attacks through carefully designed perturbations. The domain-aware explainability module depends on accurately specified physical constraint intervals; misconfigured constraints can introduce blind spots where violations go unattributed. The ARLM mitigation module was trained entirely in simulation, and policy effectiveness may degrade when deployed in real plants where network latency, sensor drift, and physical equipment dynamics differ from simulated conditions.

Regarding the real-world applicability of the autonomous mitigation capability, we emphasise that full autonomy should be treated as an aspirational endpoint rather than a default deployment mode. Several practical factors may constrain or preclude autonomous operation in specific contexts. Safety-critical installations governed by IEC 61511 or IEC 62443 standards may require certified human-in-the-loop approval before any actuator-affecting action is executed, meaning actions beyond alerting and enhanced monitoring (m_0–m_2) would require operator confirmation. Environments subject to strict regulatory oversight — such as nuclear facilities, pharmaceutical manufacturing, or water utilities — may face compliance barriers that prohibit unsanctioned automated interventions. Legacy CPS with limited API access may not expose the programmatic interfaces needed for the agent to execute isolation or shutdown commands. We therefore recommend a phased deployment model: in Phase 1, the ARLM operates in advisory mode, generating recommended actions for human approval and accumulating real-environment experience; in Phase 2, autonomy is extended

to low-severity actions (m_0–m_2) once the advisory phase demonstrates alignment with operator judgement; and in Phase 3, higher-severity actions may be authorised under pre-agreed operational conditions and with appropriate safety interlocks in place. This incremental approach allows organisations to calibrate their trust in the system and fulfil regulatory obligations while progressively realising the response-time benefits reported in this work.

Future Directions. This work can be extended in three principal directions. Federated learning would allow organisations to collaboratively build defences without disclosing sensitive operational data. Multimodal fusion, incorporating network traffic, system logs, and physical security feeds alongside sensor telemetry, would reduce the blind spots affecting any single detection modality. Most ambitiously, causal reasoning models that reason about the propagation of manipulations through physical and cyber pathways would enable attack attribution through first-principles inference, supporting more precise intervention decisions and attack-path prediction.

7. Conclusion

Protecting cyber–physical systems requires purpose-built solutions that address their unique characteristics: deterministic timing, physical consequences, heterogeneous protocols, and real-time constraints. This paper introduced XAI-CPS to meet that need.

The hybrid TAC-LSTM-GNN architecture processes temporal and spatial patterns simultaneously, detecting coordinated attacks that manipulate multiple sensors over extended periods. Across four benchmark datasets spanning water treatment, water distribution, and Industrial IoT domains, XAI-CPS achieved a 94.7% average F1-score while reducing false positive rates by 37.5%.

Rather than treating explainability as an afterthought, the domain-aware LRP mechanism integrates physical constraints directly into attribution calculations. Expert agreement with the resulting explanations reached 89.3%, supporting rapid triage decisions. The ARLM module closes the loop through reinforcement-learned response policies that balance security against operational continuity, reducing mean time to mitigate by 67% while maintaining availability above 99.2%.

These results demonstrate that sophisticated AI can serve critical infrastructure security without sacrificing interpretability or operational safety. As CPS expand across energy, transportation, and healthcare sectors, XAI-CPS offers a foundation for meeting emerging challenges through domain-aware detection, human-aligned explanations, and adaptive response. Future work will extend to federated settings, additional data modalities, and causal understanding of attack propagation.

CRediT authorship contribution statement

Iftikhar Rasheed: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Conceptualization. **Hala Mostafa:** Writing – original draft, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis. **Ghulam Mohayudin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R137), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Data availability

Data will be made available on request.

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